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 EPI 207 Assignment 2
 RQ: Among California adults 18-29, is low income (<200% FPL)
     associated with current use of any non-cigarette
     tobacco/nicotine product (incl. e-cigarettes)?
 Data: CHIS 2021-2025 adult, pooled
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LIBNAME epi "~/EPI207/Assignment 2";
OPTIONS NOFMTERR;

/*Stack 2021-2025; pooled weights (80 x 5 = 400 replicates)*/
DATA all;
  SET epi.adult2021 (IN=a)
      epi.adult2022 (IN=b)
      epi.adult2023 (IN=c)
      epi.adult2024 (IN=d)
      epi.adult2025 (IN=e);

  ARRAY orig{80} rakedw1-rakedw80;
  ARRAY fnw{400} fnwgt1-fnwgt400;

  fnwgt0 = rakedw0 / 5;

  DO i = 1 TO 400;
    fnw{i} = rakedw0 / 5;
  END;

  IF a THEN offset = 0;
  ELSE IF b THEN offset = 80;
  ELSE IF c THEN offset = 160;
  ELSE IF d THEN offset = 240;
  ELSE IF e THEN offset = 320;

  DO i = 1 TO 80;
    fnw{i + offset} = orig{i} / 5;
  END;

  year = 2021 + offset/80;

  DROP i offset;
RUN;

/*Restrict to population: age 18-29*/
DATA ya;
  SET all;
  WHERE SRAGE_P1 IN (18, 26);
RUN;

/*Recodes*/
DATA ya_recode;
  SET ya;

  /*Exposure - low income (<200% FPL)*/
  IF POVLL IN (1,2) THEN POVLL_2CAT = 1;
  ELSE IF POVLL IN (3,4) THEN POVLL_2CAT = 2;
  ELSE POVLL_2CAT = .;

  /*Outcome - any current non-cig tobacco/nicotine use, incl. e-vape*/
  IF      AC135_P1 IN (2,3) OR  /*Smokeless tobacco*/
          AC137_P1 IN (2,3) OR  /*Cigarillo/little cigar*/
          AC139_P1 IN (2,3) OR  /*Big cigar*/
          AC141_P1 IN (2,3) OR  /*Hookah*/
          ESMKCUR = 1           /*Current e-cig use*/
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  THEN Noncigpro_recode = 1;
  ELSE IF AC135_P1 IN (1,-1) AND
          AC137_P1 IN (1,-1) AND
          AC139_P1 IN (1,-1) AND
          AC141_P1 IN (1,-1) AND
          ESMKCUR = 2
  THEN Noncigpro_recode = 2;
  ELSE Noncigpro_recode = .;

  /*Race/ethnicity*/
  IF RACEDF_P1 = 6 THEN RACEDF_4cat = 1;
  ELSE IF RACEDF_P1 = 5 THEN RACEDF_4cat = 2;
  ELSE IF RACEDF_P1 = 1 THEN RACEDF_4cat = 3;
  ELSE IF RACEDF_P1 IN (2,3,4,7,8) THEN RACEDF_4cat = 4;
  ELSE RACEDF_4cat = .;

  /*Education*/
  IF AHEDC_P1 IN (1,2,3) THEN AHEDC_3cat = 1;
  ELSE IF AHEDC_P1 IN (4,5,6) THEN AHEDC_3cat = 2;
  ELSE IF AHEDC_P1 IN (7,8,9) THEN AHEDC_3cat = 3;
  ELSE AHEDC_3cat = .;

  /*ACEs*/
  IF ACE_A = 1 THEN ACE_3cat = 1;
  ELSE IF ACE_A IN (2,3,4) THEN ACE_3cat = 2;
  ELSE IF ACE_A = 5 THEN ACE_3cat = 3;
  ELSE ACE_3cat = .;

  /*Serious psychological distress (mediator; Table 1 only)*/
  IF DSTRS30 = 1 THEN spd = 1;
  ELSE IF DSTRS30 = 2 THEN spd = 2;
  ELSE spd = .;

  /*Occupation (Table 1 only)*/
  IF OCCMAIN2 IN (1:4) THEN occ_5cat = 1;
  ELSE IF OCCMAIN2 = 5 THEN occ_5cat = 2;
  ELSE IF OCCMAIN2 IN (6,7) THEN occ_5cat = 3;
  ELSE IF OCCMAIN2 IN (8:13) THEN occ_5cat = 4;
  ELSE IF OCCMAIN2 IN (-1,14) THEN occ_5cat = 5;
  ELSE occ_5cat = .;

  LABEL POVLL_2CAT       = "Low-income status (1=<200% FPL, 2=200%+ FPL)"
        Noncigpro_recode = "Non-cigarette nicotine product use incl. e-vape (1=Yes, 2=No)"
        RACEDF_4cat      = "Race/ethnicity"
        AHEDC_3cat       = "Education"
        ACE_3cat         = "Adverse childhood experiences"
        spd              = "Serious psychological distress"
        occ_5cat         = "Occupation"
        year             = "Survey year";
RUN;

/*Formats*/
PROC FORMAT;
  VALUE yesnofmt
        1 = "Yes"
        2 = "No";
  VALUE agefmt
        18 = "18-25"
        26 = "26-29";
  VALUE sexfmt
        1 = "Male"
        2 = "Female";
  VALUE racefmt
        1 = "NH White"
        2 = "NH Black"
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        3 = "Latino"
        4 = "NH Other";
  VALUE educfmt
        1 = "HS/GED or less"
        2 = "Some college"
        3 = "College or higher";
  VALUE ace3fmt
        1 = "0 ACEs"
        2 = "1-3 ACEs"
        3 = "4+ ACEs";
  VALUE ruralurbfmt
        1 = "Urban"
        2 = "Rural";
  VALUE fplfmt
        1 = "0-199% FPL"
        2 = "200% OR ABOVE FPL";
  VALUE occfmt
        1 = "Management/professional"
        2 = "Service"
        3 = "Sales/office"
        4 = "Construction/production/transport"
        5 = "Not working";
RUN;

/*Checks: recodes vs. originals, and missingness by year*/
PROC FREQ DATA=ya_recode;
  TABLES POVLL*POVLL_2CAT
         RACEDF_P1*RACEDF_4cat
         AHEDC_P1*AHEDC_3cat
         ACE_A*ACE_3cat
         DSTRS30*spd
         OCCMAIN2*occ_5cat
         AC135_P1*AC137_P1*AC139_P1*AC141_P1*ESMKCUR*Noncigpro_recode
         / LIST MISSING;
  TABLES year*(Noncigpro_recode ACE_3cat spd POVLL_2CAT) / MISSING NOROW NOCOL NOPERCENT;
RUN;

/*Table 1 variables and formats*/
%LET t1vars = Noncigpro_recode SRAGE_P1 SRSEX RACEDF_4cat AHEDC_3cat
              ACE_3cat UR_CLRT2 occ_5cat spd year;
%LET t1fmts = POVLL_2CAT fplfmt. Noncigpro_recode yesnofmt. SRAGE_P1 agefmt.
              SRSEX sexfmt. RACEDF_4cat racefmt. AHEDC_3cat educfmt.
              ACE_3cat ace3fmt. UR_CLRT2 ruralurbfmt. occ_5cat occfmt. spd yesnofmt.;

/*Table 1: overall*/
PROC SURVEYFREQ DATA=ya_recode VARMETHOD=JACKKNIFE;
  WEIGHT fnwgt0;
  REPWEIGHTS fnwgt1-fnwgt400 / JKCOEFS=1;
  TABLES &t1vars / NOWT;
  FORMAT &t1fmts;
RUN;

/*Table 1: by exposure*/
PROC SURVEYFREQ DATA=ya_recode VARMETHOD=JACKKNIFE;
  WEIGHT fnwgt0;
  REPWEIGHTS fnwgt1-fnwgt400 / JKCOEFS=1;
  TABLES POVLL_2CAT*(&t1vars) / ROW NOWT;
  FORMAT &t1fmts;
RUN;

/* TABLE 2: Logistic regression
NEEDS TO BE UPDATED*/

/*Table 2: Crude OR*/
PROC SURVEYLOGISTIC DATA=ya_recode VARMETHOD=JACKKNIFE;
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  WEIGHT fnwgt0;
  REPWEIGHTS fnwgt1-fnwgt400 / JKCOEFS=1;
  CLASS POVLL_2CAT (REF='2') / PARAM=REF;
  MODEL Noncigpro_recode (EVENT='1') = POVLL_2CAT;
RUN;

/*Table 2: Adjusted OR*/
PROC SURVEYLOGISTIC DATA=ya_recode VARMETHOD=JACKKNIFE;
  WEIGHT fnwgt0;
  REPWEIGHTS fnwgt1-fnwgt400 / JKCOEFS=1;
  CLASS POVLL_2CAT  (REF='2')
        SRAGE_P1    (REF='18')
        SRSEX       (REF='1')
        RACEDF_4cat (REF='1')
        AHEDC_3cat  (REF='3')
        ACE_3cat    (REF='1')
        UR_CLRT2    (REF='1')
        year        (REF='2021')
        / PARAM=REF;
  MODEL Noncigpro_recode (EVENT='1') =
        POVLL_2CAT SRAGE_P1 SRSEX RACEDF_4cat AHEDC_3cat ACE_3cat UR_CLRT2 year;
RUN;
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